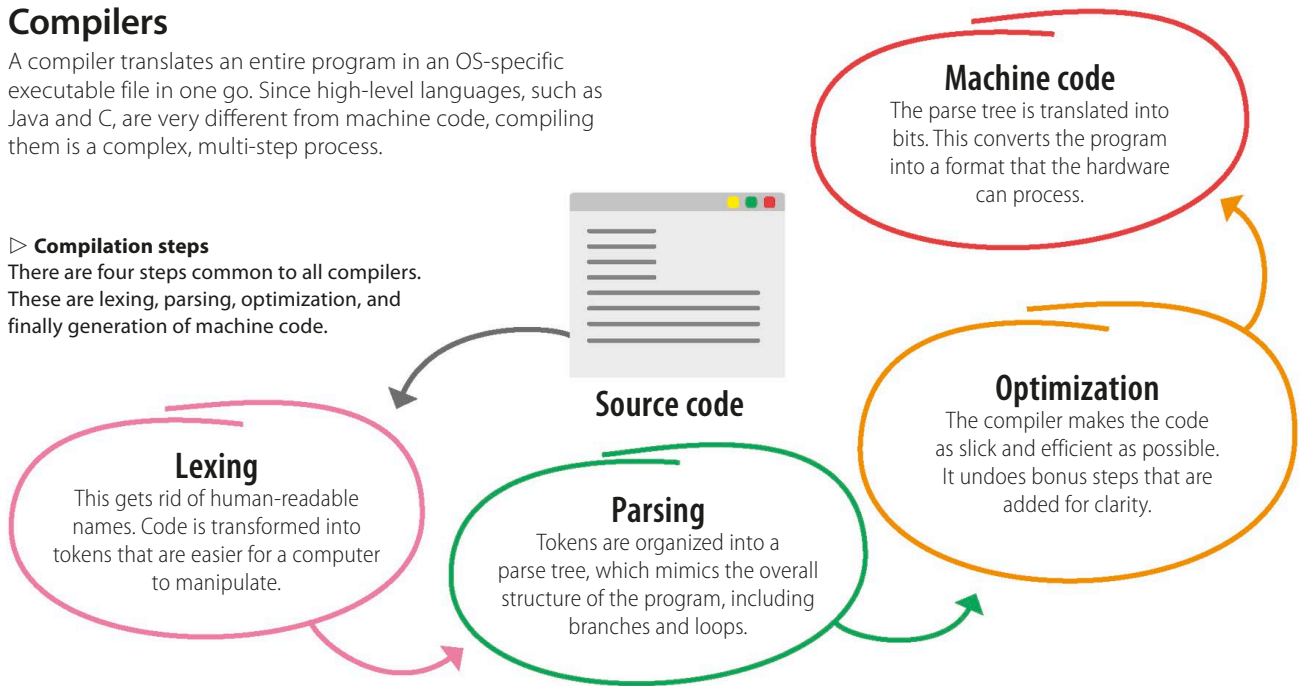


Compilers

A compiler translates an entire program in an OS-specific executable file in one go. Since high-level languages, such as Java and C, are very different from machine code, compiling them is a complex, multi-step process.

► Compilation steps

There are four steps common to all compilers. These are lexing, parsing, optimization, and finally generation of machine code.



IN DEPTH

Compilation

Compiled programs are the norm, except when browsing the internet.

Pros

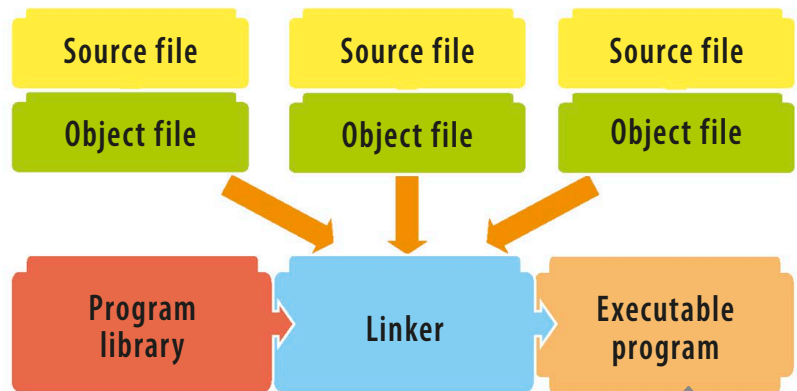
- Since programmers work with high-level code, they can work faster and make bigger, more complicated programs.
- There is less potential for mistakes.
- Once compiled, a program can be run at any time.
- Faster than interpreters.

Cons

- Compiled code is generally slower and bulkier than code written in assembly.
- Compiler errors can be vague and unhelpful. It's not always easy to find the errors in the program.
- A compiler must be written for each OS.

Linking

A program's code is typically spread out over many source code files, called object code, each of which is compiled separately. This makes it easier to change code, since modifying a single file doesn't require recompiling the entire program. However, this means that there's an extra step in the translation process: linking all of the object code together into a single program.



△ Libraries

Linkers take objects from a collection called a library. Usually, one or more system libraries are linked in by default.

Once linked, the program is ready to run.